

→ then click ▼ **red triangle** on that report's title — no JMP path, hand-computed Reject H_0 when $\mathbf{p \leq \alpha}$

PART 1 INFERENCE FOR ONE MEAN everything starts at Analyze > Distribution		
QUESTION	JMP PATH	WHERE THE ANSWER IS
Open file · fix a column type	File > Open > the .jmp file • right-click column head > Column Info > Modeling	Quantitative columns must read Continuous
Categorical or quantitative?	— read the problem	Averageable number = quantitative · group label = categorical
Define the parameter	— write a sentence	" μ = true mean [variable + units] for all [population]"
Shape, $\bar{x}$, s , n	Analyze > Distribution > Y, Columns = variable > OK	Summary Statistics: Mean, Std Dev, N, Std Err Mean
Sampling distribution of $\bar{x}$	— hand-computed	Mean = μ · SD = $\sigma/\sqrt{n}$ · Normal if pop. normal or $n \geq 30$
$P(\bar{x} > a)$	Cols > New Column > Formula > Probability > Normal Distribution($a, \mu, \sigma/\sqrt{n}$)	Gives $P(\bar{x} < a)$ — subtract from 1 for ">"
State H_0 and H_1	— read the claim's wording	$H_0: \mu = \mu_0$ · less < · greater > · different $\neq$
t test (σ not given)	Analyze > Distribution > Y = variable > OK > ▼variable > Test Mean > type μ_0 > OK	Test Mean > t Test row: Prob> t · Prob>t · Prob<t
z test (σ given)	same, but also fill "Enter True Standard Deviation to do z-test rather than t test"	Same panel, now a z Test row
Only summary stats given	— hand: $t = (\bar{x} - \mu_0) / (s/\sqrt{n})$, $df = n - 1$	s² is a variance — take $\sqrt{}$ first. Homes: $t = -1.32$, $p \approx .19$ $t = -1.03$ (below → negative), $df = n - 1$
"1.03 std errors below μ_0 "	— read the sentence	
Check the t-test conditions	— argue from the problem	Indep: random, <10% of pop. · Normal: stated, or $n \geq 30$
Is normality reasonable?	▼variable > Continuous Fit > Normal > ▼Fitted Normal > Goodness of Fit • ▼variable > Normal Quantile Plot	Shapiro-Wilk Prob<W above $\alpha \rightarrow$ normality reasonable
Confidence interval for μ	Analyze > Distribution > OK • other levels: ▼variable > Confidence Interval > 0.99	Summary Statistics: Upper / Lower 95% Mean
Interpret p · state the conclusion	— wording	$p \leq \alpha$ reject · $p > \alpha$ fail to reject . Never "accept H_0 "
Type I / Type II in context	— wording	I: reject a true H_0 (α) · II: keep a false H_0 (β)

PART 2 POWER AND TYPE II ERROR work steps 1→4 in order — almost all by hand		
QUESTION	JMP PATH	WHERE THE ANSWER IS
1. For what $\bar{x}$ do we reject?	— hand: $\bar{x}_c = \mu_0 \mp z\alpha \cdot \sigma/\sqrt{n}$ (two-tailed: $\mu_0 \pm z\alpha/2 \cdot \sigma/\sqrt{n}$)	Fertilizer: $50 - 1.645(3.5) = \mathbf{44.24}$
2. $\beta = P(\text{Type II error})$	— hand: $z = (\bar{x}_c - \mu_{\text{true}}) / (\sigma/\sqrt{n})$	$\beta = P(\bar{x} \text{ in the } \mathbf{\text{fail-to-reject}} \text{ region}) \rightarrow \mathbf{.3613}$
3. Power of the test	— hand: $\text{Power} = 1 - \beta$	Fertilizer at $\mu_{\text{true}} = 43 \rightarrow \mathbf{.6387}$
4. Power straight from JMP	DOE > Design Diagnostics > Sample Size and Power > One Sample Mean > Continue	Fill Alpha, Std Dev, Difference to Detect, n — leave Power blank
How big a sample do I need?	same screen — leave <i>Sample Size</i> blank and type the Power you want	The Sample Size box fills in
Which phrase means which?	— wording	"detecting this difference" = power · "failing to reject" = β
Sketch the graphs	— two normal curves, one vertical line at $\bar{x}_c$ through both	Top at μ_0 : tail = α · bottom at μ_{true} : fail side = β
Which error? What raises power?	— concept	Rejected $\rightarrow$ I · else II . Power $\uparrow$ with $n \uparrow$, $\alpha \uparrow$, effect $\uparrow$, $\sigma \downarrow$

PART 3 TWO PROPORTIONS independent = different people · paired = same people twice		
QUESTION	JMP PATH	WHERE THE ANSWER IS
Type in a summarized 2×2 table	File > New > Data Table > columns Group (Nom), Outcome (Nom), Count (Cont) > one row per cell	e.g. (Help, Accepted, 222)
Analyze that table	Analyze > Fit Y by X > Y = Outcome > X = Group > <i>Freq = Count</i> > OK	Contingency Table + Tests . No Freq $\rightarrow$ only 4 rows analyzed
Explanatory / response	— both are categorical here	Group $\rightarrow$ X, Factor · outcome $\rightarrow$ Y, Response
Point estimate of $\hat{p}_1 - \hat{p}_2$	▼Contingency Table > Row %	Cell + its own row total . .4311 − .2895 = .1416
H_0 and H_1	— read the claim	$H_0: p_1 - p_2 = 0$ · improves > · reduces < · differs $\neq$
Check the assumptions	— independence between and within groups	$n_1\hat{p}_1, n_1\hat{q}_1, n_2\hat{p}_2, n_2\hat{q}_2 \geq 5 = \mathbf{\text{all four cells} \geq 5}$
Test statistic and p-value	▼Contingency Analysis > Two Sample Test for Proportions	Prob> z · Prob>z · Prob<z. Pearson ChiSquare = same 2-sided test
CI for $p_1 - p_2$	same <i>Two Sample Test for Proportions</i> panel	Beside the test. Excludes 0 $\rightarrow$ agrees with rejecting H_0
Paired: label the table first	— a, d = ties (no change) · b, c = discordant (changed)	Smoking: a = 20, b = 5, c = 16, d = 9, n = 50
Paired: H_0 , ties, n' , T	— $H_0: p_{\text{b}} = p_{\text{c}}$ · ties = a+d · $n' = b+c$ · T = count in H_1 's direction	Smoking: ties 29 , n' 21 , T 16
Paired: p-value (exact binomial)	Cols > New Column > Formula > Discrete Probability > Binomial Distribution(.5, n' , k)	$T \sim \text{Bin}(n', .5)$. $1 - \text{BinDist}(.5, 21, 15) = \mathbf{.0133}$
Paired: McNemar from JMP	Analyze > Fit Y by X > Y = After, X = Before, Freq = Count > OK > ▼ > Agreement	Test of symmetry = McNemar — two-sided only
Paired: CI for $p_{\text{b}} - p_{\text{c}}$	— hand: $(b-c)/n \pm z\alpha/2\sqrt{[(a+d)(b+c) + 4bc] / n^3}$	Smoking (−.389, −.051) · contains 0 $\rightarrow$ no difference

THE MATH BEHIND THOSE NUMBERS			
ONE MEAN	POWER	PROPORTIONS	CRITICAL VALUES
$z = (\bar{x} - \mu_0) / (\sigma/\sqrt{n})$ $t = (\bar{x} - \mu_0) / (s/\sqrt{n})$ $df=n-1$ $CI \bar{x} \pm t(\alpha/2, n-1) \cdot s/\sqrt{n}$ $CI \bar{x} \pm z\alpha/2 \cdot \sigma/\sqrt{n}$ (σ known)	1 $\bar{x}_c = \mu_0 \mp z\alpha \cdot \sigma/\sqrt{n}$ 2 $z = (\bar{x}_c - \mu_{\text{true}}) / (\sigma/\sqrt{n})$ 3 $\beta = P(\text{fail to reject} \mu_{\text{true}})$ 4 $\text{Power} = 1 - \beta$	$\hat{p} = (x_1 + x_2) / (n_1 + n_2)$ pooled $z = (\hat{p}_1 - \hat{p}_2) / \sqrt{[\hat{p}\hat{q}(1/n_1 + 1/n_2)]}$ $CI (\hat{p}_1 - \hat{p}_2) \pm z\alpha/2\sqrt{[\hat{p}_1\hat{q}_1/n_1 + \hat{p}_2\hat{q}_2/n_2]}$ paired $CI (b-c)/n \pm z\alpha/2\sqrt{[n'/n^3]}$	1-tail z .10 1.282 .05 1.645 .025 1.960 .01 2.326 2-tail 90% 1.645 95% 1.960 98% 2.326 99% 2.576

TRAPS THAT COST POINTS			
Prob> t is two-sided. Halve for one-tailed H_1 only if the sign agrees; else $p > .5$. Mean vs. Indiv. A singular noun ("a truck", "a school") wants the wider <i>Indiv.</i> Blank Y. A row added for a prediction needs an empty Y or it moves the line.	Freq box. A count table without Count in <i>Freq</i> is read as 4 observations. Paired $\neq$ two samples. Same people twice $\rightarrow$ drop ties, binomial test. Center Polynomials. Uncheck it before reading raw quadratic coefficients.	"Holding all others constant" in every multiple-regression interpretation. Units. x in thousands: 34,000 is $x = 34$, not 34000. df: $n-1$ one mean · $n-2$ simple slope · $n-k-1$ multiple coefficient.	Never "accept H_0," never claim causation, never put $\bar{x}$ in a hypothesis. Row %. $\hat{p}$ divides by that group's own row total. R² can't compare models of different sizes — use <i>RSquare Adj.</i>

PART 4 CORRELATION & SIMPLE LINEAR REGRESSION			Fit Y by X for plots and r · Fit Model for everything else
QUESTION	JMP PATH	WHERE THE ANSWER IS	
Explanatory vs. response	– "predict A using B" → y = A, x = B	y → Y, Response · x → X, Factor	
Scatterplot · comment on it	Analyze > Fit Y by X > Y, Response = response > X, Factor = factor > OK	Direction, form, strength, outliers. Curved → try a quadratic	
Add the line to the plot	▼Bivariate Fit > Fit Line	Equation printed as text above Summary of Fit	
Correlation r	Analyze > Multivariate Methods > Multivariate > both variables in Y, Columns > OK	Correlations , off-diagonal · or ±√RSquare, sign of slope	
Equation of the least squares line	Analyze > Fit Model > Y = response > select x > Add > Run	Parameter Estimates > Estimate : Intercept = b ₀ , x row = b ₁	
Predict y when x = ____	– substitute into $\hat{y} = b_0 + b_1x$	Units! x in thousands → "34,000 employees" is x = 34	
Interpret slope · intercept	– wording	Per +1 x-unit, predicted y changes b ₁ · b ₀ = $\hat{y}$ at x = 0	
Save residuals & predicted values	▼Response y > Save Columns > Residuals / Predicted Values / Studentized Residuals	New columns at the far right of the data table	
Residual histogram + NQ plot	Analyze > Distribution > Y = Residual > OK > ▼Residual > Normal Quantile Plot	Then Continuous Fit > Normal > Goodness of Fit → Shapiro-Wilk	
Residual plot · constant variance?	▼Response y > Row Diagnostics > Plot Residual by Predicted	Band = OK · arch = wrong form · funnel = variance not constant	
Is independence reasonable?	– argue from the study design	Random sample of separate units; careful with year-by-year data	
Is x a useful predictor?	Fit Model > Run > Parameter Estimates > the x row	t Ratio, Prob> t , df = n – 2. Halve p if one-sided	
Test β ₁ = a value other than 0	– hand: t = (b ₁ – c) / Std Error, or see whether the CI contains c	Muscle Mass: does the 99% CI for the slope contain –2?	
Proportion of variation explained	Summary of Fit	RSquare × 100%	
95% CI for the slope	right-click inside Parameter Estimates > Columns > Lower 95% + Upper 95%	Read on the x row, not Intercept	
99% or 98% CI for the slope	▼Response y > Set Alpha Level > 0.01 (or Other → 0.02), then re-read those columns	If locked at 95%: b₁ ± t(α/2, n–2) · Std Error	
CI for the mean response at x	add row: x filled, Y blank > Fit Model > Run > Save Columns > Mean Confidence Interval	Lower / Upper 95% Mean , on the row you added	
Prediction interval, one new case	same setup > ▼Response y > Save Columns > Indiv Confidence Interval	Lower / Upper 95% Indiv — always the wider one	
Draw the bands on the plot	▼Linear Fit > Confid Curves Fit • ▼Linear Fit > Confid Curves Indiv	Graph only — gives no numbers	
Unusual point · new variable	click the point > Rows > Exclude/Unexclude > re-Run • Cols > New Column > Formula	Extra votes = actual – predicted = that row's residual	
PART 5 MULTIPLE & POLYNOMIAL REGRESSION			same report — more rows in Parameter Estimates
QUESTION	JMP PATH	WHERE THE ANSWER IS	
All predictors linear? intercorrelated?	Analyze > Multivariate Methods > Multivariate > response + all predictors in Y, Columns > OK	Correlations + Scatterplot Matrix . r > .8 = concern	
Fit the model · get the equation	Analyze > Fit Model > Y = response > select all predictors > Add > Run	Parameter Estimates > Estimate , one row per predictor	
Interpret one coefficient	– wording	per +1 unit of x _k , holding all other predictors constant	
Is the model useful?	Fit Model > Run > Analysis of Variance	F Ratio, Prob > F . H ₀ : all β = 0	
Adjusting for the rest, is ____ useful?	Fit Model > Run > Parameter Estimates > that predictor's row	t Ratio, Prob> t , df = n – k – 1	
CI for one coefficient	right-click Parameter Estimates > Columns > Lower 95% + Upper 95%	Other levels: ▼Response y > Set Alpha Level first	
Check the conditions	▼Response y > Save Columns > Residuals • ▼ > Row Diagnostics > Plot Residual by Predicted	Then Distribution on the residuals → Shapiro-Wilk	
Multicollinearity, numerically	right-click Parameter Estimates > Columns > VIF	VIF > 5 warning, > 10 a real problem	
Which model is best?	Summary of Fit, on each model you fit	RSquare Adj — highest wins. Plain R ² always rises	
List every possible model	Fit Model > add all > Personality = Stepwise > Run > ▼ > All Possible Models	RSquare, RSquare Adj , Cp, AICc, BIC	
Backward model selection	Fit Model > add all > Stepwise > Run > Backward > P-value Threshold > Prob to Leave .1 > Go	Current Estimates : a check under Entered = kept	
Full report for the final model	in the Stepwise window > Make Model > Run	The normal Fit Model report	
Which variable goes out first?	– read the full model's p-values	The largest p-value , if > 0.1. Remove one at a time	
Predict at given predictor values	add a row with those x's, Y blank > Run > Save Columns > Predicted Values	Or ▼Response y > Factor Profiling > Profiler	
Fit a quadratic — quick route	Analyze > Fit Y by X > Y, X > OK > ▼Bivariate Fit > Fit Polynomial > 2, quadratic	Polynomial Fit Degree=2 ; uncentered equation above it	
Fit a quadratic — full route	Analyze > Fit Model > Y > select x > Macros > Polynomial to Degree (2) > Run	Adds an x*x row. Use this route for intervals	
Get the raw b ₀ , b ₁ , b ₂	▼Response y > uncheck <i>Center Polynomials</i> > Run again	JMP centers by default, printing (x–$\bar{x}$)·(x–$\bar{x}$)	
Is the quadratic model useful?	Parameter Estimates > the x*x row • or the Effect Tests table	Prob> t on the squared term. H ₀ : β ₂ = 0, not the overall F	
Change in y from x=2 to x=3 · CI / PI	– hand: b ₁ (x ₂ –x ₁) + b ₂ (x ₂ ² –x ₁ ²) • add row (x, Y blank) > Save Columns > Mean / Indiv CI	0→1 = b₁+b₂ · 2→3 = b₁+5b₂ — not equal, that's the point	

THE MATH BEHIND THOSE NUMBERS — REGRESSION		
LEAST SQUARES $S_{xx} = \sum (x - \bar{x})^2$ $S_{yy} = \sum (y - \bar{y})^2$ $S_{xy} = \sum (x - \bar{x})(y - \bar{y})$ $b_1 = S_{xy}/S_{xx}$ $b_0 = \bar{y} - b_1\bar{x}$ $r = S_{xy}/\sqrt{(S_{xx} \cdot S_{yy})}$ $R^2 = r^2$	TESTS & DF slope $t = b_1/SE$ $df = n-2$ vs. value c $t = (b_1 - c)/SE$ $df = n-2$ coefficient $t = b_k/SE$ $df = n-k-1$ overall F $df = k$ and $n-k-1$	INTERVALS & R²ADJ slope $b_1 \pm t(\alpha/2, n-2) \cdot SE(b_1)$ coef $b_k \pm t(\alpha/2, n-k-1) \cdot SE(b_k)$ $R^2_{adj} = 1 - (1-R^2)(n-1)/(n-k-1)$ quadratic $\hat{y} = b_0 + b_1x + b_2x^2$